

Designing a Responsible Multi-Agent Safety Architecture for *Autonomous Vehicle Decision Systems*

Avneet Kaur
Anish Srinivasan

"Responsibility isn't a layer you add. It's a shape you choose."

The Accountability Gap

When AVs fail, today's architectures cannot tell us where the failure was, or who should have caught it.

PERCEPTION + ACCOUNTABILITY

Uber ATG, Tempe

2018 · fatality

WHAT HAPPENED

The vehicle detected Elaine Herzberg roughly six seconds before impact. The perception system reclassified her between vehicle, other, and bicycle. Each reclassification reset her tracking history.

WHY IT MATTERS

Automatic emergency braking had been disabled by Uber engineers to prevent erratic behavior. The safety driver was distracted. NTSB found no clear locus of decision responsibility.

No agent owned the uncertainty signal.

HUMAN HANDOFF

Tesla Autopilot

2016 to ongoing · multiple fatal crashes

WHAT HAPPENED

Repeated crashes into stationary emergency vehicles. NHTSA opened a formal investigation in 2021, expanded its scope, and issued multiple recalls. The pattern is consistent across incidents.

WHY IT MATTERS

The system assumes the driver is monitoring. Human factors research shows drivers cannot maintain vigilance over reliable automation. Often, handoff requests arrive at the moment vigilance is lowest.

Handoff treated as a feature, not a failure mode.

In both cases, we cannot fully reconstruct what the system thought, or who was supposed to catch the failure. The accountability gap is structural, not accidental.

What We're Building (and What We're Not)



THIS IS

A design brief

A reference architecture, worked scenarios, and a traceability template. Software and architecture-level only.



THIS IS NOT

A simulator or empirical evaluation

We do not run a simulator, measure safety outcomes, or claim empirical performance. Behavioral claims are design-time, not measured.



WE CONTRIBUTE

Component-level RAI mapping

A mapping of NIST AI RMF and EU AI Act requirements onto a real-time multi-agent AV pipeline, plus a concrete log template that demonstrates what accountability looks like in operation.

Framework Anchoring

Four frameworks ground the design. Each does distinct, load-bearing work.



NIST AI RMF

(2023)

CONTRIBUTES

Map / Measure / Manage scaffolding for per-agent risk analysis.

INFORMS

Agents 2, 3, 6



EU AI Act

(2024)

CONTRIBUTES

AV systems are classified as high-risk AI. Mandates oversight, logging, audit.

INFORMS

Agents 5, 6



Jobin et al.

(2019)

CONTRIBUTES

Cross-cutting principles distilled from 84 AI ethics guidelines.

INFORMS

Used as audit rubric (Slide 10)



Owen et al.

Resp. Innovation

CONTRIBUTES

Anticipatory framing: imagine failure modes before deployment.

INFORMS

Drives edge cases (Slide 8)

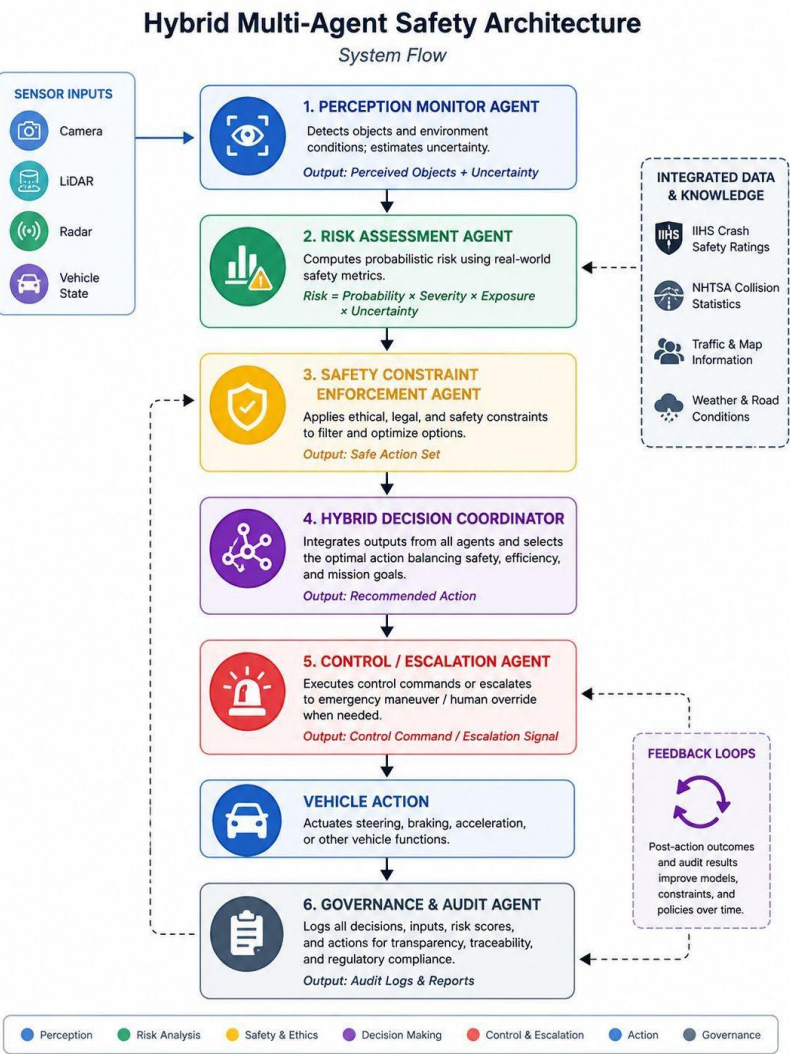


Plus AV-specific and human-factors literature

NTSB Tempe report · NHTSA Tesla investigations · Casner & Hutchins on automation handoff · Varshney & Alemzadeh on ML safety

The Hybrid Multi-Agent Architecture

Specialized agents for analysis · Explicit coordination for accountability · Dedicated governance for traceability



Why hybrid? Two failure modes to avoid.

- Centralized**
One model decides everything. Fast but opaque, with no observability into intermediate reasoning.
- Decentralized**
Highly modular but no clear arbiter. Agent disagreements have no resolution path.
- Our hybrid**
Specialized agents (1-3) for analysis, explicit Coordinator (4) for decisions, dedicated Governance (6) for audit.

Why 6 agents, not 4?

Our original proposal had four agents but didn't specify where final decisions actually got made or how they'd be audited. We added the Coordinator (Agent 4) and Governance (Agent 6) because those are exactly the gaps responsible design has to close.

Agent Decision Boundaries: Analysis Agents

What each agent is allowed to decide, what it isn't, and what triggers escalation. (1 of 2)

1  AGENT	Perception Monitor DECIDES Object classification + uncertainty score. Flags degraded sensor conditions. DOES NOT DECIDE Whether action is needed. Risk magnitude. ESCALATES WHEN Uncertainty exceeds threshold τ_p, or sensor degradation flagged (low light, occlusion, glare).
2  AGENT	Risk Assessment DECIDES Probabilistic risk score: $\text{Risk} = P \times \text{Severity} \times \text{Exposure} \times \text{Uncertainty Factor}$. DOES NOT DECIDE Action choice. Constraint definitions. ESCALATES WHEN Adjusted risk exceeds critical threshold τ_r.
3  AGENT	Safety Constraint Enforcement DECIDES Permissible (safe) action set after applying ethical, legal, and safety constraints. DOES NOT DECIDE Final action. Risk weighting. ESCALATES WHEN Permissible action set is empty (all options violate something).

→ continues on next slide (Coordinator, Control, Governance)

Agent Decision Boundaries: Decision & Audit Agents

Where decisions get made, executed, and recorded. (2 of 2)

4  AGENT	Hybrid Decision Coordinator DECIDES Final action choice from the safe action set, balancing risk, efficiency, mission goals. DOES NOT DECIDE Constraint definitions. Risk model parameters.	ACCOUNTABILITY LOCUS ESCALATES WHEN Confidence below threshold, or unresolvable conflict between agent recommendations.
5  AGENT	Control / Escalation DECIDES Execute selected action OR initiate handoff to human operator. DOES NOT DECIDE Strategic action choice. Constraint definitions.	EXECUTE OR HAND OFF ESCALATES WHEN ODD violation, low coordinator confidence, empty safe set, sustained sensor degradation.
6  AGENT	Governance & Audit DECIDES Nothing in real-time. Records every input, output, and decision across all agents. DOES NOT DECIDE Any control decision. Any agent's output.	TRACEABILITY BY DESIGN ESCALATES WHEN Anomaly pattern detected post-hoc → flagged for regulator review.

Agent 4 is the single accountability locus. Agent 6 makes traceability a designed function

Edge Cases & Expected Behaviors

Three named edge cases tied to real failures. Each stresses a different part of the architecture.

TEMPE-STYLE

Low-visibility pedestrian

STRESSOR

High perception uncertainty in poor lighting

STRESSES

Perception → Risk pathway

EXPECTED OBSERVABLE BEHAVIOR

Perception flags HIGH uncertainty → Risk inflated by uncertainty factor → Constraint admits braking only → Coordinator selects conservative action → Full audit trace logged.

Uncertainty is propagated, not smoothed away.

TESLA-STYLE

Stationary obstacle at highway speed

STRESSOR

Late detection at high closing speed; potential takeover required

STRESSES

Coordinator → Control/Escalation pathway

EXPECTED OBSERVABLE BEHAVIOR

Coordinator selects max safe deceleration as the system's fallback → Escalation request issued early with predicted time-to-impact → Driver-monitoring state logged → If handoff fails, fallback continues to execute, fully traced.

Handoff is logged and backstopped, not assumed successful.

GRACEFUL DEGRADATION

Sensor degradation in heavy rain

STRESSOR

Camera confidence drops mid-trip; LiDAR preturns degraded by precipitation.

STRESSES

Perception → Constraint feedback loop

EXPECTED OBSERVABLE BEHAVIOR

Perception flags degraded mode → Constraint agent narrows allowable speed range → Governance logs mode transition → System operates within reduced ODD or escalates if outside it.

Degradation produces visible behavioral change.

Traceability is demonstrated, not asserted

One scenario. One log. The full decision lineage made auditable.

SCENARIO

Low-visibility pedestrian

Tempe-style scenario

Vehicle approaches a pedestrian in low light. Perception confidence is low. The architecture must propagate uncertainty as risk and produce a conservative, auditable response.

WHAT THIS LOG DEMONSTRATES

Transparency every agent's I/O is visible.

Safety uncertainty inflates risk, not dampens it.

Accountability Coordinator records explicit rationale.

Oversight escalation state is logged either way.

Privacy log fields minimized, no raw sensor data.

Traceability single audit_id retrieves the full chain.

GOVERNANCE & AUDIT LOG • audit_id #A2891

[t=0.00s] **Perception:** pedestrian, conf=0.42, uncertainty=HIGH, flag=low_light

[t=0.04s] **Risk:** P=0.18, S=critical, Exposure=2.1s,
Adjusted=0.61 (u_factor=1.4)

[t=0.08s] **Constraint:** {brake_hard} ∈ safe_set;
{evade_L, evade_R} ∉ safe_set
(oncoming, shoulder_obstructed)

[t=0.12s] **Coordinator:** action=brake_hard,
rationale=conservative_under_uncertainty

[t=0.14s] **Control:** executing brake_hard, no_escalation

[t=0.16s] **Governance:** trace_logged, audit_id=#A2891

An investigator retrieves audit_id #A2891 and replays the entire decision chain. That is how traceability is demonstrated.

Mapping RAI Principles to Architecture

Each principle has a mechanism, and a named limitation.

Transparency	MECHANISM Per-agent inputs and outputs logged by Agent 6. KNOWN LIMITATION <i>Logs are not explanations; interpretation still requires translation for non-technical audiences.</i>	Accountability	MECHANISM Coordinator (Agent 4) is the single, designed decision locus. KNOWN LIMITATION <i>Structural only. Legal liability remains contested; this informs but does not resolve it.</i>
Human Oversight	MECHANISM Escalation paths and ODD checks in Agent 5. Coordinator holds a conservative fallback while handoff is in flight. Driver-monitoring state logged. KNOWN LIMITATION <i>Does not solve unsafe handoff. The fallback reduces severity but cannot guarantee a safe outcome.</i>	Safety	MECHANISM Risk × Uncertainty Factor in Agent 2; constraint filtering in Agent 3. KNOWN LIMITATION <i>Uncertainty estimates are themselves uncertain (calibration is an open problem).</i>
Fairness	MECHANISM Out of scope at the architecture level. Flagged as a known gap. KNOWN LIMITATION <i>Requires dataset- and training-level intervention not addressed by this design.</i>	Privacy	MECHANISM Audit logs minimized to decision-relevant fields; no raw sensor data, no PII. KNOWN LIMITATION <i>Tension between auditability and data minimization; more logging means more retention.</i>

The 'Known Limitation' column is what makes this responsible analysis instead of marketing.

What This Design Does Not Fix

TECHNICAL

- Six-agent coordination has real latency cost. Real-time AV systems operate at 10-100ms cycles; our pipeline budget is tight.
- Uncertainty estimates are themselves uncertain. Calibration of confidence scores remains an open ML safety problem.
- Adversarial inputs could potentially deadlock agents (e.g., perception flips repeatedly, constraint set oscillates).

SOCIO-TECHNICAL

- Human takeover (Agent 5) inherits the Tesla failure mode. We do not solve unsafe handoff. What we change: handoffs become investigable (full trace including driver-monitoring state) and the Coordinator holds a conservative fallback action while handoff is in flight.
- Operators may game escalation thresholds to reduce nuisance alerts, turning a safety mechanism into a bypass.
- Drivers' attention degrades within minutes of reliable automation. Casner & Hutchins document this consistently.

GOVERNANCE

- Logs are useful only if regulators have authority and expertise to audit them. Current US framework is fragmented.
- EU AI Act provides stronger audit teeth, but cross-border AV operation creates jurisdictional gaps.
- Logging at this fidelity raises retention, privacy, and liability questions we do not resolve.

Our Contribution

Three things this design brief offers -

01

Component-level RAI mapping

Most Responsible-AI work happens at the organization or product level. We push it down to the architecture: each agent has principles it instantiates and limitations it inherits. This is where regulation has to actually meet code.

02

A named tension in the literature

Modularity helps engineering but fragments accountability, unless coordination and governance are themselves designed components. The hybrid form responds directly to this tension.

03

A traceability template

A concrete log format that demonstrates what accountability looks like in operation. Detailed enough to be critiqued, formatted enough to be implemented, honest enough about its limits.

Responsibility isn't a layer you add. It's a shape you choose.

Thank you.

Questions and discussion welcome.

Avneet Kaur
Anish Srinivasan

"Responsibility isn't a layer you add. It's a shape you choose."

George Mason University